

Title	Airline Passenger & Route Analysis
Skill Takeaway From This Project	Data Cleaning & Preprocessing Using Excel SQL & Database Management Python Data Analysis Power BI & Dashboard Development
Domain	Transportation

Problem Statement:

An airline wants to evaluate route profitability, load factor, and on-time performance. The analysis should track load factor by route and month, flag underperforming flights, study on-time performance and revenue per route, and examine how delays relate to route, season, and passenger satisfaction. Present the findings through clear visual summaries and an interactive dashboard covering route performance, load-factor trends, and delay analysis.

Business Use Cases

1. **Route Profitability Analysis** – Identify high- and low-revenue routes to support route planning and resource allocation.
2. **Load Factor Optimization** – Monitor passenger occupancy to improve seat utilization and flight efficiency.
3. **Delay Management** – Identify frequently delayed routes, airlines, and seasons to improve on-time performance.
4. **Passenger Satisfaction Improvement** – Analyze how delays and flight conditions affect customer satisfaction.
5. **Operational Decision-Making** – Use interactive dashboards to support scheduling, route planning, and overall airline performance decisions.

Approach:

Phase 1: Python – Data Cleaning, Programming & Analysis

1. Airline Dataset Profiling

Load all airline-related datasets using Pandas and examine dataset dimensions, column structures, data types, missing values, duplicate records, and descriptive statistics.

2. Data Quality & Cleaning

Identify and handle missing values, duplicate records, inconsistent categorical values, and incorrect data types across the airline datasets.

3. Flight Performance Classification

Classify flights based on load factor using conditional logic to identify low-, medium-, and high-performing flights.

4. Flight Delay Assessment

Analyze flight records to identify and count flights experiencing significant delays, particularly flights delayed by more than **30 minutes**.

5. Staff Salary Analysis

Use a for loop to iterate through staff records and identify and count employees whose salary exceeds a specified threshold.

6. Experienced Staff Identification

Use a while loop to process staff records and identify the first few employees with experience above a specified number of years.

7. Route Performance Analysis

Use Pandas groupby() to calculate total revenue, passenger count, average load factor, and average delay for each route.

8. Airline Performance Analysis

Evaluate airline performance using average delay, total revenue, passenger volume, and average load factor, and identify the best- and worst-performing airlines.

9. Integrated Airline Data Analysis

Merge Flights, Passengers, Routes, Airlines, and Airports datasets to create an integrated dataset for cross-dimensional analysis of routes, airlines, delays, and passenger satisfaction.

10. Statistical & Exploratory Analysis

Calculate mean, median, standard deviation, and IQR for key numerical variables such as revenue, load factor, flight duration, ticket price, and passenger satisfaction. Use correlation analysis and Matplotlib/Seaborn visualizations to identify patterns, relationships, and unusual values across the airline data.

Phase 2: SQL – Database Management and Querying

1. Create Tables Based on ERD

- Use CREATE TABLE statements to create the airline database tables with appropriate data types, primary keys, foreign keys, and constraints.

2. Import CSV Files into SQL

- Import the cleaned airline CSV files into MySQL using LOAD DATA or the MySQL Workbench import feature.

3. Passenger and Flight Details Using JOINS

- Join the Passengers and Flights tables to display passenger details along with flight number, airline name, and destination airport.

4. Passenger Ticket and Travel Details

- Use JOINS to display passenger name, ticket price, travel class, and flight status.

5. Highest Delay Flight Analysis

- Find the flight with the highest delay and display its flight number, airline, route, and delay duration.

6. Route Travel Time Analysis

- Identify the routes with the highest and lowest average travel time.

7. Airline Delay Performance

- Calculate the average delay for each airline and identify the airline with the highest average delay.

8. Load Factor by Route

- Calculate the average load factor for each route to identify high- and low-performing routes.

9. Monthly Load Factor Analysis

- Analyze the average load factor for each route by month to identify changes in flight occupancy.

10. Revenue by Route

- Calculate the total revenue generated by each route and rank the routes from highest to lowest revenue.

11. Underperforming Flight Analysis

- Identify flights with a load factor below 60% and delay greater than 30 minutes.

12. On-Time Performance Analysis

- Calculate the number and percentage of delayed flights for each airline.

13. Delayed vs On-Time Passenger Satisfaction

- Compare the average passenger satisfaction score between delayed and on-time flights.

14. Route, Delay and Satisfaction Analysis

- Analyze the average passenger satisfaction for each route based on flight status to understand how delays affect passenger experience.

15. Top 5 Route Performance

- Find the top 5 routes by total revenue and display their average load factor and average delay.

Phase 3: Power BI – Interactive Dashboard & Visualization

1. Data Import and Model Creation

- Import the airline datasets into Power BI and create relationships between the Flights, Passengers, Routes, Airlines, Airports, and Date tables.

2. KPI Cards – Overall Performance

- Create KPI cards showing Total Flights, Total Passengers, Total Revenue, Average Load Factor, and Average Delay.

3. Revenue by Route

- Create a bar chart showing total revenue by route and identify the highest-revenue routes.

4. Load Factor by Route

- Create a bar chart comparing the average load factor across routes.

5. Monthly Load Factor Trend

- Create a line chart showing how the average load factor changes month by month.

6. Revenue Trend by Month

- Create a line or area chart showing monthly revenue trends.

7. On-Time vs Delayed Flights

- Create a donut or column chart showing the number and percentage of on-time, delayed, and cancelled flights.

8. Delay Analysis by Airline

- Create a bar chart comparing average delay minutes across airlines.

9. Passenger Satisfaction Analysis

- Compare average passenger satisfaction across different flight statuses, airlines, or travel classes.

10. Delay vs Passenger Satisfaction

- Create a scatter chart to visualize the relationship between delay duration and passenger satisfaction.

11. Underperforming Flight Analysis

- Create a DAX measure or calculated column to identify flights with low load factor and high delay, and display them in a table or chart.

12. Seasonal Performance Analysis

- Analyze revenue, load factor, and delays by season using a slicer or suitable visual.

13. Interactive Slicers and Drill-Down

- Add slicers for Airline, Route, Month, Season, Flight Status, and Travel Class, and enable drill-down for route and time-based analysis.

14. Create the Final Interactive Airline Dashboard

- Combine KPIs, route performance, load-factor trends, revenue, delay analysis, and passenger satisfaction into an interactive dashboard with appropriate filters and visual formatting.

Results

- Identified high- and low-performing routes based on revenue and load factor.
- Analyzed flight delays by route, airline, month, and season.
- Identified underperforming flights with low load factor and high delays.
- Examined the impact of delays on passenger satisfaction.
- Developed an interactive Power BI dashboard for airline performance monitoring.

Project Evaluation Metrics

- Total Revenue
- Average Load Factor (%)
- Average Delay (Minutes)
- On-Time Flight Percentage
- Average Passenger Satisfaction Score

Technical Tags

Python | Pandas | NumPy | Matplotlib | Seaborn | MySQL | SQL | Power BI | DAX | Data Cleaning | Data Analysis | Data Visualization | Dashboard

Dataset

- Dataset Name: Airline Passenger & Route Analysis Dataset
- Format: CSV files
- Number of Tables: 8
- Records: 18,000+ across all tables
- Key Variables: Flight ID, Passenger ID, Airline, Route, Flight Date, Scheduled Departure, Actual Departure, Passengers, Seat Capacity, Load Factor, Delay Minutes, Flight Status, Base Fare, Revenue, Travel Class, Satisfaction Score, Airport, Staff, Aircraft, Season.